BTS-GANs: Method for Imputing Missing Values to Improve Predictive Modelling Power in Climate Change

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INTRODUCTION

- The Hanford Site historically released contaminants to the subsurface during production of plutonium.
- Hexavalent chromium (Cr(VI)) is one of the contaminants released to the subsurface.
- We need additional methods to forecast Cr(VI) concentrations in groundwater based on local environmental conditions, like local temperature, to optimize environmental cleanup operations.

PROBLEM STATEMENT

- Could changes in local temperature impact subsurface conditions with subsequent impacts to Cr(VI) contaminant fate and transport?
- The problem will be investigated based on the objectives below:

1. To order the dataset collected in different time periods resulting in an inconsistent timestamp.
2. To impute missing values/sequences for improving the performance of model forecast.
3. Testing a long-short-term memory (LSTM) model on imputed data.

SORTING THE DATASET

- We consider Cr(VI) and local temperature parameters.
- As the temperature changes locally, precipitation patterns, evaporation rates, groundwater recharge, etc., can influence the transport, fate, and distribution of contaminants in groundwater.
- The data collection time is inconsistent and hampers the model's forecast performance.
- The sample collection date is converted to a Unix timestamp and sorted. For example,

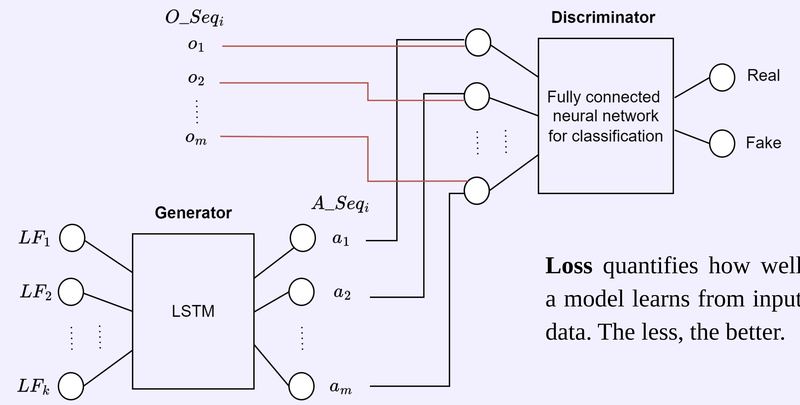
Samp_Date_Time	Value	Unix Time
03-01-2002 12:36	24.4	1010083000
17-04-1993 11:40	28.5	735064840
09-01-2007 13:13	23.8	1168370020

Samp_Date_Time	Value	Unix Time
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IMPUTING MISSING VALUES AND SEQUENCES

- Since the timestamps for these two parameters differ, we extract samples biweekly.
- However, there are many missing values.
- Missing values are imputed by generative adversarial networks (GAN) with batching time-series (BTS).
- Time-series data are divided into multiple smaller batches of length 8, to train GAN.
- Following the training of a GAN, it is prompted to generate an artificial batch, which is used to compare each value with its actual batch (except the missing values). The cumulative difference is noted.

Proposed architecture



Loss quantifies how well a model learns from input data. The less, the better.

Figure 1. Training GAN.

- After several prompts, an artificial batch that generates a small difference is used to obtain the missing values.
- This process is executed for each parameter.

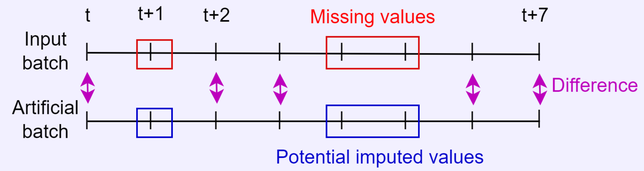


Figure 2. Imputing missing values.

RESULTS

Experimental settings

- Cr(VI) and temperature are the input parameters and Cr(VI) is the output parameter. The aim is to forecast Cr(VI) in the Hanford 100 area.
- The input and output sequence length is 4.
- Performance is assessed using LSTM model on original (without missing values) and imputed data (using different methods).

Performance analysis

- Baseline imputation models include Mean (mean of the neighbour values), KNN (neighbour similarity), and Regression (ML-based) methods.
- 10% of the values from the original dataset were marked at random as missing.
- On data imputed by Mean, KNN, and Regression, LSTM forecast model loss increased by 5.3%, 44.6%, and 65.38%, respectively from the model trained with the original dataset (without missing values).

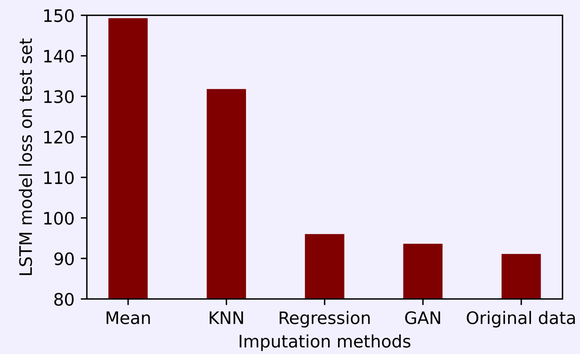


Figure 3. Model loss for imputation methods on the test set.

- In contrast, LSTM on the data imputed by BTS-GANs result in only 2.7% loss increase.

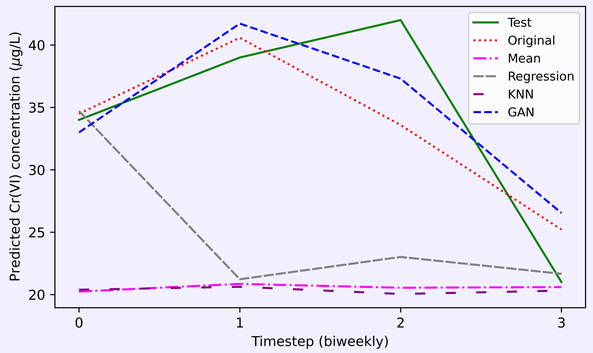


Figure 4. Cr(VI) concentration forecast for the test set.

CONCLUSION

We propose BTS-GANs to impute missing values and improve Cr(VI) forecast performance.

REFERENCES

1. J. Kim, et al., "*A Survey of Missing Data Imputation Using Generative Adversarial Networks*," ICAIIC, Fukuoka, Japan, 2020, pp. 454-456.
2. "*PNNL database*," (https://phoenix.pnnl.gov/phoenix/apps/gallery/index.html)
3. "*Hanford Meteorological Station*," https://www.hanford.gov/page.cfm/HMS.

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BENEFITS

- The proposed method can help the DoE improve forecast accuracy when missing values are present in the dataset.
- BTS-GANs can be used to treat missing values for other environmental parameters.